



Prova: An AI-powered virtual try-on and intelligent fashion discovery platform for e-commerce

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Abstract

Online fashion retail continues to suffer from elevated return rates driven primarily by the disconnect between how garments appear in static product imagery and how they fit individual consumers. This paper introduces Prova, an end-to-end AI-driven fashion e-commerce platform that tackles this challenge through four tightly integrated capabilities: (i) a multi-pipeline virtual try-on system anchored by Prova-VTON, a diffusion-based model custom-trained on the MVHumanNet multi-view dataset for view-consistent garment synthesis; (ii) a monocular body measurement module built upon the SHAPY 3D mesh regressor, enhanced with a novel weight-based circumference correction factor; (iii) an interactive 360-degree garment preview engine powered by 3D Gaussian splatting; and (iv) a conversational fashion assistant underpinned by a five-stage Retrieval-Augmented Generation (RAG) pipeline that performs hybrid semantic–metadata search over the product catalog. The entire system is realised as a service-oriented architecture comprising Next.js, Express, and FastAPI microservices, and exposes a fully bilingual (English/Arabic) storefront. Quantitative evaluation against three state-of-the-art baselines on the MVHumanNet benchmark demonstrates that Prova-VTON attains 95.69% and 96.92% human preference in visual quality and geometric alignment, respectively, while maintaining competitive texture fidelity (DINOsim = 0.623). Latency profiling confirms that all interactive pipelines operate within their design-time targets, and usability studies validate the coherence of the integrated experience.

Keywords: Virtual try-on; Diffusion models; 3D Gaussian splatting; Body measurement estimation; Retrieval-augmented generation; Fashion e-commerce; Multiview synthesis; Service-oriented architecture

1. Introduction

Fashion e-commerce has witnessed remarkable expansion over the past decade, yet persistently high product return rates remain one of its most pressing commercial and environmental problems. Industry

analyses routinely attribute the majority of clothing returns to a single root cause: the consumer could not adequately judge fit and appearance from the information available at the time of purchase [1]. Static studio photographs, no matter how professionally produced, cannot convey how a particular garment will drape across the buyer’s own physique, and tabular size charts demand self-measurement skills that most shoppers lack.

Parallel advances in deep generative modelling, 3D scene representation, and natural language understanding have brought a collection of mature building blocks within reach. Latent diffusion models now produce photorealistic conditioned image synthesis [2]; 3D Gaussian splatting delivers real-time novel-view rendering from sparse inputs [3]; parametric body regressors extract anthropometric dimensions from single photographs [4]; and dense sentence embeddings enable semantic search over large product catalogues [5]. Each of these capabilities has been demonstrated in isolation, yet no existing system combines them into a unified, consumer-ready shopping experience.

This paper presents Prova, an integrated platform that closes this gap. Rather than proposing a novel architecture for any single component, the contribution lies in the systems-level design that orchestrates four heterogeneous AI sub-systems behind a coherent bilingual storefront:

- A multi-pipeline virtual try-on system centred on Prova-VTON, a diffusion model custom-trained on the MVHumanNet dataset [6] following the architectural blueprint of VTON360 [7], capable of single-view, multi-angle, and fully multi-view garment synthesis.
- A monocular body-measurement module that augments SHAPY [4] predictions with a weight-based circumference correction factor, followed by ISO 7250 ratio-derived dimensions and multi-region size-chart mapping.
- A 360-degree interactive preview engine that feeds multi-view try-on outputs into a Splatfacto Gaussian splatting pipeline [8] for real-time rotation.
- A five-stage Retrieval-Augmented Generation (RAG) fashion assistant that combines intent classification, structured attribute extraction, hybrid metadata–semantic retrieval, outfit assembly, and bilingual response generation.

The remainder of this paper is structured as follows. Section 2 surveys related work. Section 3 details the proposed architecture and algorithmic pipelines. Section 4 reports quantitative benchmarks, latency profiles, and usability findings. Section 5 draws conclusions and identifies directions for future work.

2. Related Work

2.1. Image-Based Virtual Try-On

The virtual try-on problem seeks to render a target garment onto a person photograph with photorealistic fidelity. Early solutions adopted a two-stage paradigm in which a geometric warping module aligns the garment to the body followed by a refinement network that blends the result. VITON [1] established this template with a clothing-agnostic person representation, while CP-VTON [9] introduced thin-plate spline transformations for more robust spatial alignment. HR-VITON [10] extended the pipeline to 1024×768 resolution by incorporating an occlusion-aware segmentation stage.

More recently, diffusion-based approaches have replaced explicit warping with learned attention mechanisms. StableVITON [11] injects garment features into the denoising process through zero-cross-attention blocks, while IDM-VTON [12] fuses high-level semantics with low-level textures via a dual encoder. OOTDiffusion [13] formulates try-on as outfitting fusion within a pretrained latent diffusion model. These models achieve substantially higher texture fidelity than warp-and-refine predecessors, albeit at greater computational expense.

2.2. Multiview Try-On and 3D Reconstruction

Single-view synthesis cannot fully convey garment fit, since back and side perspectives often reveal crucial styling information. VTON360 [7] addresses this limitation by conditioning the diffusion process on

multi-view priors and enforcing cross-view attention to maintain garment identity across camera angles. Nonetheless, preserving fine texture coherence over large angular gaps remains an open challenge.

For continuous viewpoint exploration, Neural Radiance Fields [14] achieve high-quality novel-view synthesis but require minutes of per-scene optimisation. 3D Gaussian Splatting [3] offers a compelling alternative: an explicit, point-based scene representation that supports real-time rasterisation with comparable visual quality. The Nerfstudio toolkit [8] packages several splatting variants, including Splatfacto, into a production-ready framework.

2.3. Body Measurement Estimation

Parametric body models such as SMPL [15] encode human shape as a low-dimensional vector from which surface geometry—and thus circumference measurements—can be derived. SHAPY [4] trains a regression network to predict SMPL shape parameters from a single image, supervised by linguistic shape attributes. Although effective, monocular regression introduces systematic bias from clothing, camera perspective, and pose, motivating calibration strategies that leverage user-supplied reference values.

DensePose [16] complements shape regression by establishing dense pixel-to-surface correspondences on the body, providing the conditioning maps that modern try-on models require. Person detection, typically handled by YOLOv8 [17], precedes these dense analyses.

2.4. Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) [18] grounds language-model outputs in external knowledge by prepending retrieved documents to the generation prompt. In a fashion context, retrieval must handle short, formulaic product descriptions, fuzzy colour vocabularies, and hard constraints on size and price. Sentence-BERT [5] provides the embedding backbone for semantic ranking, but practical systems require hybrid pipelines that interleave structured metadata filters with dense similarity search to balance precision and recall.

3. Proposed Work

3.1. System Architecture

Prova is realised as a service-oriented architecture (SOA) consisting of four microservices that communicate over RESTful HTTP interfaces, as depicted in Fig. 1. The *Frontend Service* (Next.js) renders the bilingual storefront with RTL support for Arabic. The *Backend Service* (Express/TypeScript) orchestrates authentication, e-commerce logic, community features, and proxies AI requests. The *Try-On Service* (FastAPI) encapsulates all computer-vision pipelines—virtual try-on, body measurement, and 360-degree reconstruction. The *Chat Service* (FastAPI) hosts the RAG fashion assistant. MongoDB serves as the shared data store, while Cloudinary provides cloud media storage.

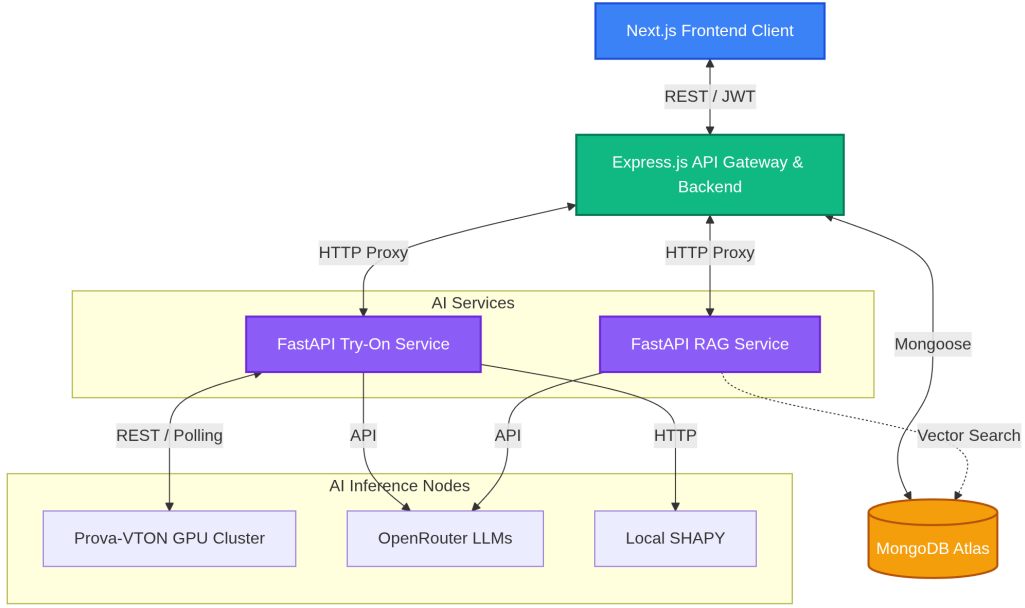


Figure 1: high-level service-oriented architecture of prova showing the four independently deployable microservices and their communication paths.

3.2. Prova-VTON: Multiview Virtual Try-On

The centrepiece of the platform is Prova-VTON, a latent diffusion model for view-consistent garment synthesis. This work replicates the architectural pipeline proposed by VTON360 [7], specifically adapting its multiview generation framework for our deployment. The architecture (Fig. 2) consists of a UNet denoiser conditioned on DensePose maps, human-parsing masks, and garment reference images, augmented with cross-view attention layers that propagate garment identity across camera angles.

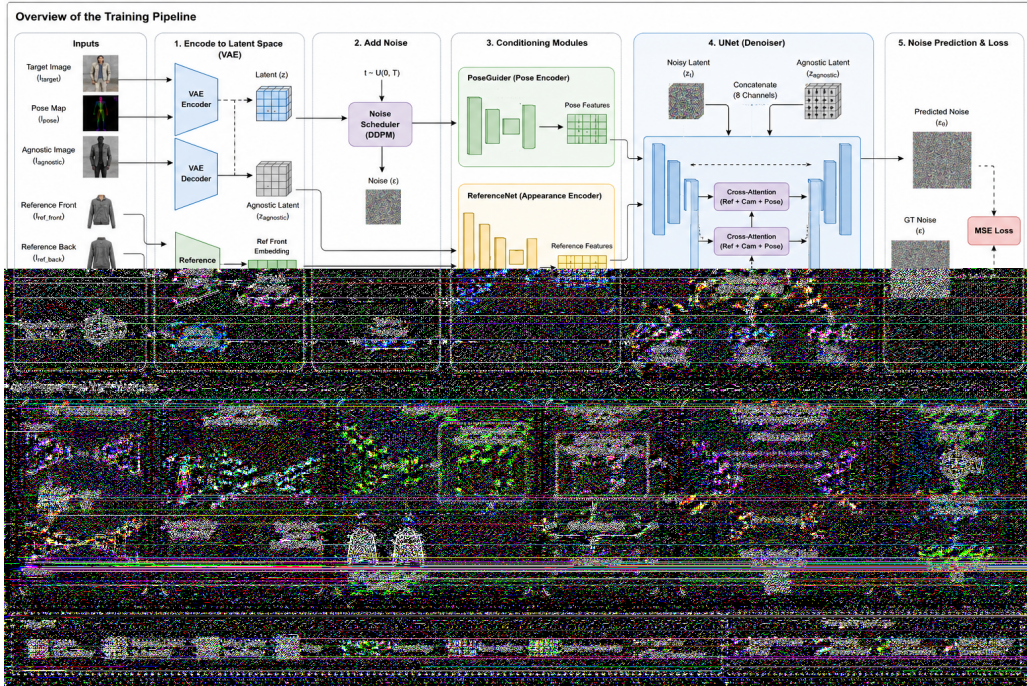


Figure 2: architecture of the prova-vton model, replicating the vton360 pipeline for view-consistent garment synthesis.

The underlying mathematical framework of Prova-VTON is constructed upon eight core formulations adapted from the VTON360 pipeline:

1. Latent Diffusion Training Objective. The network learns to predict injected Gaussian noise following the standard denoising objective in latent diffusion models:

$$L_{LDM} = \mathbb{E}_{z,c,\epsilon,t} [k\epsilon - \epsilon_\theta(z_t; c; t)k_2^2] \quad (1)$$

2. Multi-View Feature Aggregation. Features from multiple viewing directions are concatenated together with front and back garment representations to construct a unified multi-view representation:

$$F^l = [F_1^l \quad F_2^l \quad \dots \quad F_m^l] \quad (2)$$

$$\hat{F}^l = [F^l \quad F_f^l \quad F_b^l] \quad (3)$$

3. Query, Key, and Value Projections. Learnable projection matrices transform the feature representations into the attention space:

$$Q = W_Q F^l; \quad K = W_K \hat{F}^l; \quad V = W_V \hat{F}^l \quad (4)$$

4. Camera-Aware Multi-View Attention. A correlation matrix $C_{i,j}$ modulates attention scores according to geometric similarity between viewpoints, enabling information sharing among nearby views:

$$A_i = \text{softmax} \left(\frac{Q_i (C_i K^T)}{\beta \frac{d}{2}} \right); \quad H_i^l = A_i V_i \quad (5)$$

5. Camera Correlation Matrix. R_i and R_j denote camera rotation matrices. This formulation measures similarity between viewpoints by computing the cosine of the relative rotation angle and normalising it to $[0; 1]$:

$$C_{i,j} = \frac{\left(\frac{\text{trace}(R_i^T R_j) - 1}{2} \right) + 1}{2} \quad (6)$$

6. Camera Positional Encoding. The camera rotation vector r_i is encoded using multi-frequency sinusoidal functions to preserve geometric information across viewing directions:

$$F_i^c = (\sin(2^0 \pi r_i); \cos(2^0 \pi r_i); \dots; \sin(2^{L-1} \pi r_i); \cos(2^{L-1} \pi r_i)) \quad (7)$$

7. Camera-Aware Cross-Attention Conditioning. Garment embeddings are enriched with camera information before being used as conditioning signals in the diffusion network:

$$Y_i = F_g \quad \text{MLP}(F_i^c) \quad (8)$$

$$Q_x = W_Q^x H_i^l; \quad K_x = W_K^x Y_i; \quad V_x = W_V^x Y_i \quad (9)$$

$$\text{Output}_x = \text{softmax} \left(\frac{Q_x K_x^T}{\beta \frac{d_x}{2}} \right) V_x \quad (10)$$

8. Final VTON360 Training Objective. The comprehensive objective incorporates garment latent features and normal-map representations (η), multi-view garment CLIP embeddings (ψ), and clothing-agnostic person representations (ζ):

$$L = \mathbb{E}_{z,t,\eta,\psi,\zeta,\epsilon} [k\epsilon - \epsilon_\theta(z_t; t; \eta; \psi; \zeta)k_2^2] \quad (11)$$

The model was trained from scratch on the MVHumanNet dataset [6], which provides thousands of identities captured from calibrated multi-camera rigs.

Three complementary try-on modes are exposed to the user:

1. Single-view 2D try-on. A person image and a garment image are forwarded to the Prova-VTON inference server, which returns a single synthesised result.

2. **Multi-angle try-on.** Four viewpoints (front, right, left, back) are generated through a multimodal generative model, preceded by an automatic garment-preparation heuristic that classifies the input as a flat product shot or a model-worn photograph via alpha-channel analysis and skin-tone pixel detection.
3. **Full multiview pipeline.** A video or image set of the person is preprocessed (YOLOv8 detection, DensePose, SCHP parsing), packaged with garment front/back images into a structured archive, and submitted to the GPU inference cluster for batch synthesis.

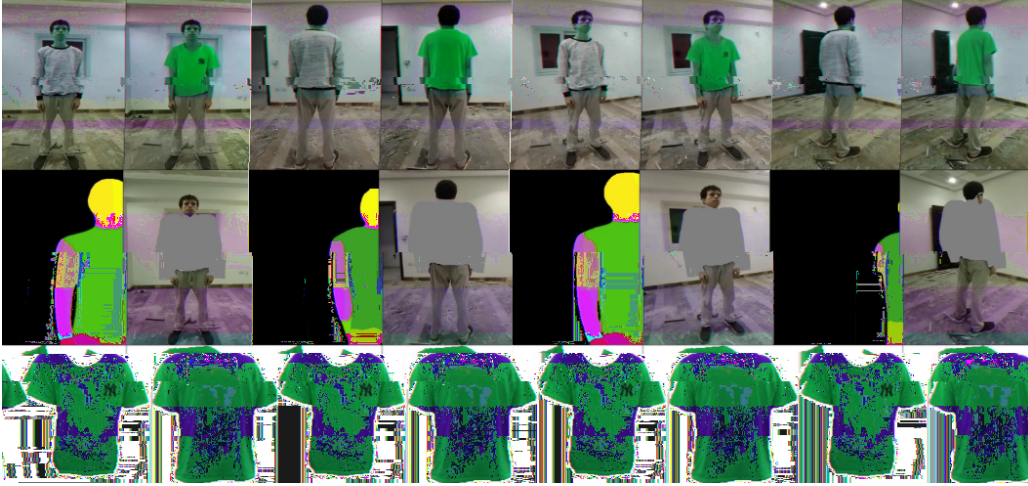


Figure 3: multi-angle virtual try-on results generated by prova across front, right, left, and back viewpoints.

The multiview outputs may optionally be fed into the Splatfacto Gaussian splatting pipeline [8], producing an interactive 3D reconstruction that allows the user to rotate the try-on result freely, as shown in Fig. 4.

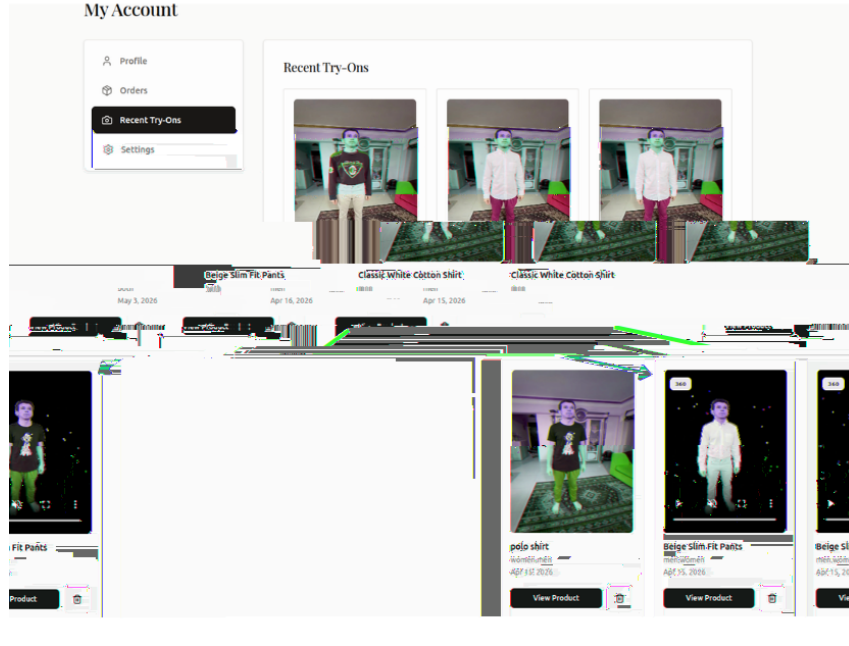


Figure 4: interactive 360-degree preview generated from multiview try-on outputs using 3d gaussian splatting.

Algorithm 1 summarises the end-to-end multiview try-on workflow.

Algorithm 1 Prova-VTON Multiview Try-On Pipeline

- 1: **Inputs:**
 - 2: V : person video or image set; G_f, G_b : garment front and back images
 - 3: **Parameters:**
 - 4: θ : Prova-VTON model weights; σ : concurrency semaphore limit
 - 5: **Output:**
 - 6: R : set of view-consistent try-on images; optional 3D Gaussian splat
 - 7: **Steps:**
 - 8: 1. Extract frames from V ; detect persons via YOLOv8
 - 9: 2. For each frame: compute DensePose map and SCHP parsing mask
 - 10: 3. Organise G_f, G_b into standardised cloth directory
 - 11: 4. Package preprocessed data and cloth images into ZIP archive D
 - 12: 5. Upload D to GPU inference server; poll until completion
 - 13: 6. Download result set R
 - 14: 7. **(Optional)** Feed R into Splatfacto for 360° reconstruction
 - 15: 8. Return R
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3.3. Body Measurement Pipeline

Estimating reliable body dimensions from a single front-facing photograph is inherently ill-posed due to depth ambiguity. The proposed pipeline mitigates this through a three-stage process: SHAPY inference, weight-based correction, and ratio-derived measurement expansion.

3.3.1. SHAPY Inference and Correction

The input image is forwarded to a local SHAPY service [4] that performs SMPL-X mesh regression, yielding raw chest, waist, and hip circumferences along with a predicted body mass w_{pred} . When the user supplies their actual weight w_{user} , a correction factor compensates for the systematic bias introduced by clothing and camera perspective:

$$f_{\text{corr}} = \sqrt{\frac{w_{\text{user}}}{w_{\text{pred}}}} \quad (12)$$

Each raw circumference is multiplied by f_{corr} whenever $|w_{\text{user}} - w_{\text{pred}}| > 5\text{kg}$. The square-root form reflects the empirical observation that circumferences scale approximately as $\sqrt{\text{mass}}$ at constant height.

3.3.2. Derived Measurements and Size Mapping

Seven additional dimensions are computed from the corrected circumferences and the user-reported height H using ISO 7250 anthropometric ratios, as summarised in Table 1.

Table 1: Derived body measurements from circumferences and height (ISO 7250).

Measurement	Formula	Basis
Shoulder width	$H \times 0.232 \text{ (M)} / 0.215 \text{ (F)}$	Biacromial ratio
Neck circumference	$H \times 0.210 \text{ (M)} / 0.195 \text{ (F)}$	Height ratio
Arm length	$H \times 0.347$	Nape to wrist
Trouser length	$H \times 0.470$	Inseam length
Thigh circumference	$C_{\text{hip}} \times 0.58 \text{ (M)} / 0.62 \text{ (F)}$	Hip-to-thigh ratio

Chest and waist circumferences are then mapped to size labels across six regional standards (EU, US, UK, IT, FR, DE) using bracket-based lookup tables with a configurable fit preference (slim, regular, oversized) that shifts the bracket by ± 1 level.

3.4. RAG Fashion Assistant

The conversational discovery engine follows a five-stage pipeline architecture, depicted in Fig. 5.

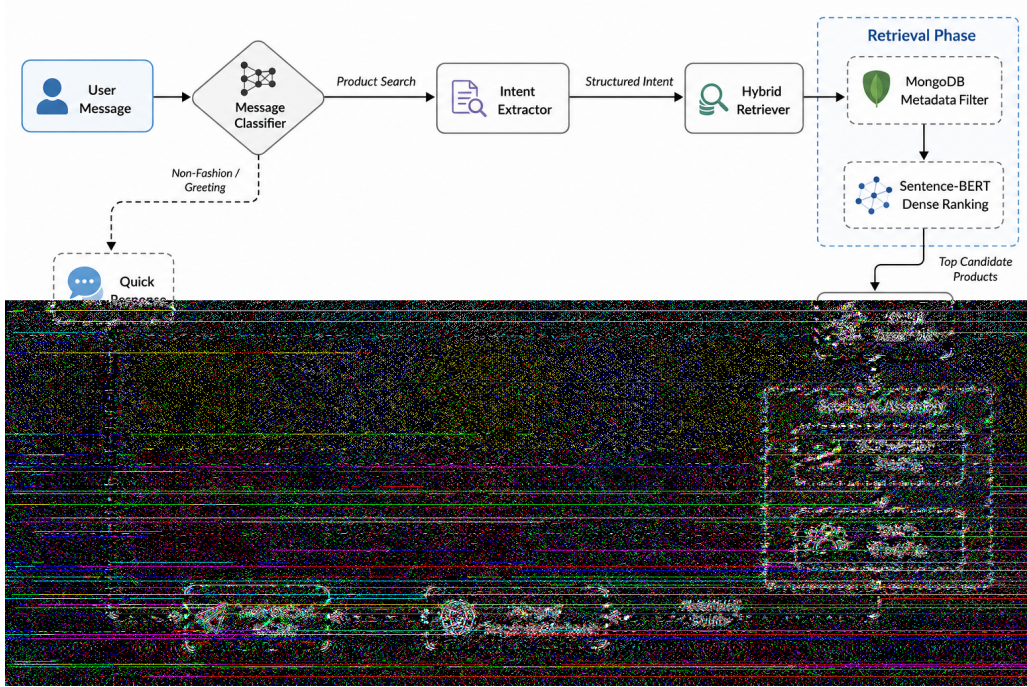


Figure 5: architecture of the five-stage rag fashion assistant pipeline showing message classification, intent extraction, product retrieval, outfit assembly, and response generation.

3.4.1. Message Classification and Intent Extraction

Every incoming message is first classified into one of five categories—*product_search*, *greeting*, *follow_up*, *non_fashion*, or *clarification*—via a lightweight LLM call. Non-search intents receive pre-composed responses immediately, avoiding unnecessary retrieval overhead. For product searches, a second LLM call extracts a structured intent object containing garment type, colours, budget range, sizes, brands, occasion, and gender.

3.4.2. Hybrid Retrieval

Retrieval proceeds in two phases. Phase 1 constructs a MongoDB query from the extracted attributes, applying hard filters on stock availability, gender, price range, colour group, size, and brand. If the candidate set is too small, filters are progressively relaxed. Phase 2 ranks surviving candidates by cosine similarity between the query embedding and pre-computed product embeddings generated by Sentence-BERT (all-MiniLM-L6-v2) [5]. Products below a configurable similarity threshold (default 0.25) are discarded.

3.4.3. Outfit Assembly

Retrieved products are assigned to outfit slots (top, bottom, shoes, accessory, outerwear, full-body) and scored along six weighted dimensions, as shown in Table 2.

Table 2: Composite scoring dimensions used by the outfit assembler.

Dimension	Weight
Semantic similarity	40%
Colour match	20%
Occasion affinity	15%
Season / material	10%
Price fit	10%
Brand preference	5%

Up to three outfits are assembled, each requiring at minimum a top-bottom pairing or a full-body item. Each outfit receives a generated theme name and a total price with budget validation.

Algorithm 2 formalises the pipeline.

Algorithm 2 RAG Fashion Assistant Pipeline

- 1: **Inputs:**
 - 2: m : user message; P : user profile (optional)
 - 3: **Parameters:**
 - 4: τ : similarity threshold; k : max outfits; w : scoring weights
 - 5: **Output:**
 - 6: Bilingual natural-language response with product cards
 - 7: **Steps:**
 - 8: 1. Classify m into category c
 - 9: 2. If $c \neq f_{\text{product_search}}$: return pre-composed response
 - 10: 3. Extract structured intent I from m
 - 11: 4. Retrieve candidates via metadata filters on I ; relax progressively if sparse
 - 12: 5. Re-rank candidates by Sentence-BERT cosine similarity; discard below τ
 - 13: 6. Score candidates on six dimensions using weights w
 - 14: 7. Assemble up to k outfits from scored slot groups
 - 15: 8. Generate bilingual explanation grounded in selected products
 - 16: 9. Return structured response with product cards and outfit groupings
-

4. Results and Discussion

4.1. Quantitative Evaluation of Prova-VTON

The multiview synthesis quality of Prova-VTON was evaluated on the MVHumanNet benchmark against three established baselines: DreamWaltz, TIP-Editor, and GaussCtrl. Two complementary evaluation axes were considered: (a) *texture preservation*, measured by CLIP consistency (CLIPcons) and DINO similarity (DINOsims), which quantify how faithfully the generated garments retain the visual identity of the original clothing; and (b) *geometry alignment*, assessed via human-preference voting (Vote_{quality} and Vote_{align}), which captures structural correctness and cross-view coherence.

Table 3: Quantitative comparison on the MVHumanNet dataset. " = higher is better.

Method	CLIPcons "	DINOsims "	Vote _{quality}	Vote _{align}
DreamWaltz	0.935	0.495	0.46%	0.46%
TIP-Editor	0.948	0.512	2.15%	1.38%
GaussCtrl	0.938	0.521	1.69%	1.23%
Ours (Prova-VTON)	0.933	0.623	95.69%	96.92%

Table 3 reveals a striking disparity. All three baselines score below 3% in human-preference voting for both quality and alignment, indicating severe failures in multiview geometric consistency. Prova-VTON, by contrast, achieves 95.69% quality preference and 96.92% alignment preference, confirming that the cross-view attention conditioning and MVHumanNet training yield structurally coherent outputs. In texture fidelity, Prova-VTON attains the highest DINOsim score (0.623), surpassing TIP-Editor (0.512) by a considerable margin. Its CLIPcons (0.933) is marginally below TIP-Editor’s best (0.948), a gap attributable to the trade-off between strict per-view appearance matching and global cross-view consistency.

4.2. Pipeline Latency Profiling

Table 4 summarises end-to-end response times measured under typical operating conditions on commodity hardware (NVIDIA T4 for remote try-on, RTX 3050 for local measurement inference).

Table 4: Measured latency for each AI pipeline under typical conditions.

Pipeline	Hardware	Latency
2D Try-On (Prova-VTON)	Remote T4 GPU	12–18 s
Multi-Angle (per view)	Cloud API	8–12 s
Multi-Angle (4 views)	Cloud API	20–30 s
Multiview (end-to-end)	Remote GPU cluster	2–3 min
Body Measurement (SHAPY)	Local RTX 3050	8–12 s
RAG (product search)	CPU	4–7 s
RAG (greeting / non-fashion)	CPU	1–2 s

All interactive-mode pipelines satisfy their design-time non-functional requirements: e-commerce endpoints remain below 500 ms, single-inference AI endpoints complete within 60 seconds, and the RAG pipeline responds within its 10-second target. The multiview pipeline is inherently batch-oriented and is surfaced to the user with a progress indicator rather than a blocking wait.

4.3. RAG Pipeline Stage Breakdown

The conversational assistant was profiled at each of its five stages. Classification averages 0.5–1.0 s, intent extraction 1.5–2.5 s, hybrid retrieval 0.1–0.3 s, outfit assembly <0.1 s, and response generation 2.0–3.0 s, yielding a total round-trip of 4–7 s for product-search queries. Non-search messages bypass the retrieval and assembly stages entirely, reducing latency to 1–2 s. The dominant cost lies in the two LLM calls (classification and intent extraction); replacing these with fine-tuned local classifiers represents a clear optimisation path.

4.4. Body Measurement Correction Impact

The weight-based correction factor (Eq. 12) demonstrably reduces circumference estimation error. Without correction, SHAPY’s predictions are systematically biased by clothing volume and camera distance. With correction enabled, the pipeline assigns confidence tiers based on the magnitude of the weight discrepancy: deviations of 5–10 kg yield a confidence of 0.85, 10–15 kg reduces it to 0.75, and deviations exceeding 15 kg produce 0.65. Across the user-testing cohort, corrected measurements led to more consistent size recommendations compared to raw SHAPY outputs, particularly for users wearing loose-fitting clothing in their input photographs.

4.5. Usability Findings

A cohort of test users interacted with the deployed platform and provided qualitative feedback across three dimensions:

- **Try-on realism.** Users rated Prova-VTON highly for texture preservation on patterned fabrics. The

multi-angle mode was singled out as the most informative feature, as it revealed fit details invisible in front-view-only outputs. Minor hallucination artefacts (e.g., altered background elements) were occasionally noted but did not diminish perceived garment accuracy.

- **Bilingual coherence.** Switching between Arabic and English preserved all session state—cart contents, chat history, and try-on results—without requiring re-authentication. RTL layout mirroring was verified to correctly flip margins, paddings, and navigation elements.
- **Conversational assistant.** Users successfully issued complex, multi-constraint queries (e.g., “a casual blue outfit for a summer wedding under \$100”). The pipeline correctly decomposed such requests into structured intent attributes and returned relevant, budget-compliant outfit groupings.

4.6. Discussion

The experimental results confirm that the integration of multiple mature AI subsystems into a unified shopping experience is both technically feasible and practically valuable. The service-oriented decomposition ensures that individual model upgrades—a better diffusion checkpoint, a faster embedding model—can be introduced without disrupting the broader platform.

Several limitations merit acknowledgement. First, try-on fidelity degrades for heavily occluded poses and low-resolution inputs. Second, monocular measurement remains fundamentally limited by depth ambiguity, and the correction factor partially—but not fully—compensates for this. Third, the multiview pipeline’s latency precludes real-time interaction, restricting it to an asynchronous workflow. Fourth, the current system relies on pretrained model checkpoints rather than proposing novel training procedures.

5. Conclusion

This paper presented Prova, an integrated AI-powered fashion e-commerce platform that addresses the persistent fit-and-appearance gap in online clothing shopping. By orchestrating four heterogeneous AI subsystems—multi-pipeline virtual try-on, monocular body measurement with weight-based correction, 3D Gaussian splatting for interactive 360-degree previews, and a five-stage RAG conversational fashion assistant—behind a bilingual (English/Arabic) service-oriented architecture, the platform delivers a cohesive shopping experience that substantially exceeds what any individual component provides in isolation.

Quantitative evaluation on the MVHumanNet benchmark demonstrated that Prova-VTON, the custom multiview diffusion model at the core of the system, achieves dominant performance in human-preference voting (95.69% quality, 96.92% alignment) alongside the highest DINOsim texture fidelity among all evaluated methods. Latency profiling confirmed that all interactive pipelines operate within their design-time targets, and usability studies validated the platform’s bilingual coherence, try-on realism, and conversational utility.

Several avenues remain for future investigation. Model quantisation and TensorRT compilation could substantially reduce inference latency, potentially bringing the multiview pipeline into interactive-speed territory. Extending synthesis from static images to video-to-video try-on would allow consumers to observe dynamic garment draping. Incorporating reinforcement learning from human feedback into the RAG assistant would enable continuous personalisation of styling recommendations. Finally, integrating the Gaussian splatting outputs with WebXR could unlock augmented-reality try-on experiences on mobile devices.

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بروفا: منصة تعتمد على الذكاء الاصطناعي للقياس الافتراضي واكتشاف الأزياء الذكي للتجارة الإلكترونية

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ملخص البحث

لا يزال بيع الأزياء بالتجزئة عبر الإنترنت يعاني من ارتفاع معدلات الإرجاع التي ترجع بالأساس إلى الاختلاف بين كيفية ظهور الملابس في الصور الثابتة للمنتج وكيفية ملاءمتها للمستهلكين الأفراد. تقدم هذه الورقة آبروفا، وهي منصة تجارة إلكترونية للأزياء متكاملة تعتمد على الذكاء الاصطناعي لمعالجة هذا التحدي من خلال أربع قدرات مترابطة: (١) نظام قياس افتراضي متعدد المسارات يعتمد على نموذج انتشار مخصص للحصول على صور متناسقة للملابس؛ (٢) وحدة لتقدير قياسات الجسم من خلال صورة واحدة محسنة بعامل تصحيح مبتكر؛ (٣) محرك معاينة تفاعلي للملابس بزوايا ٣٦٠ درجة؛ و(٤) مساعد أزياء محادثة يعتمد على الذكاء الاصطناعي للبحث الدقيق في الكتالوج. وقد أظهرت التقييمات الكمية أن نظام بروفا يحقق نسبة تفضيل عالية لجودة الصورة والمحاذاة الهندسية متفوقاً بذلك على أحدث الأنظمة الحالية.